

Self-adaptive seam detection framework for unmanned structural steel welding robots in unstructured environments

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ABSTRACT

Existing vision-based seam detection frameworks for structural steel welding robots perform well in static, predefined workspaces but struggle in dynamic, unstructured real production environments. Therefore, a self-adaptive seam detection framework is proposed that enables welding robots to interpret the actual welding environment through an estimation of the target workpiece pose. An enhanced pose estimation algorithm is developed to generate a reliable workpiece pose estimation based on solely one RGB image. Based on this real-time workpiece pose, the system autonomously determined the necessary robot movements toward an optimal position and orientation for subsequent high-precision structured light sensor measurements. The seam extraction and welding trajectory planning were then completed through an automated process. Experimental results demonstrate that the proposed framework not only enables cost-effective, fully automated weld seam detection in dynamic unstructured environments, but also achieves 72 % higher efficiency than conventional methods while eliminating human intervention.

1. Introduction

Welding robots have exponentially boosted productivity and significantly reduced risks to human welders in the construction industry [1]. The robotic system detects the weld seams of interest before it performs the welding process [2]. Effective and accurate weld seam detection is fundamental to robotic welding [3]. An autonomous robotic welding process is largely dependent on the intelligence and automation of seam detection techniques.

In common industrial practices, weld seams are detected by the robotic system through visual sensing techniques [4]. Existing vision sensors can be categorized into passive vision and active vision [5]. Passive vision, which relies on ambient or existing light sources to capture visual data [6], is low-cost [7] and enables more comprehensive information acquisition [8]. By applying passive vision techniques, welding robots managed to identify the target weld seam within complex backgrounds [9]. However, the lack of high-precision 3D data and accurate depth information makes passive vision difficult to apply in practical production [10].

In comparison, active vision techniques facilitate point cloud acquisition by actively interacting with the environment through

projecting structured light or signals onto the target [11]. The capability of capturing high-quality 3D data makes active vision the prevailing option for steel welding robots [12]. Cheng et al [13] presented a line structured light-based sensing system to recognize weld seams and used RTMDet-nano to reduce noise. Xu et al [14] used the FPLDN network to detect feature points in line laser images for 3D weld seam reconstruction. Geng et al. [15] exploited a 3D surface scanning industrial camera to acquire point clouds and designed specific rules to extract weld seams based on geometric features.

However, current active vision-based seam detection techniques are constrained to static and structured environments due to their limited measurement range, which is not consistent with the dynamic, unstructured environment that exists in actual steel workpiece fabrication [16]. As shown in Fig. 1, the configuration of the steel workpiece demonstrates significant uncertainty in real production, preventing active vision sensors with a narrow field of view from capturing point cloud data from the target area. As a consequence, substantial human intervention is typically needed to ensure successful seam detection. Tavares et al. [17] proposed placing the workpiece on customized equipment to limit the uncertainty of the workpiece pose to meet the requirement of active vision, but introducing additional appliances not

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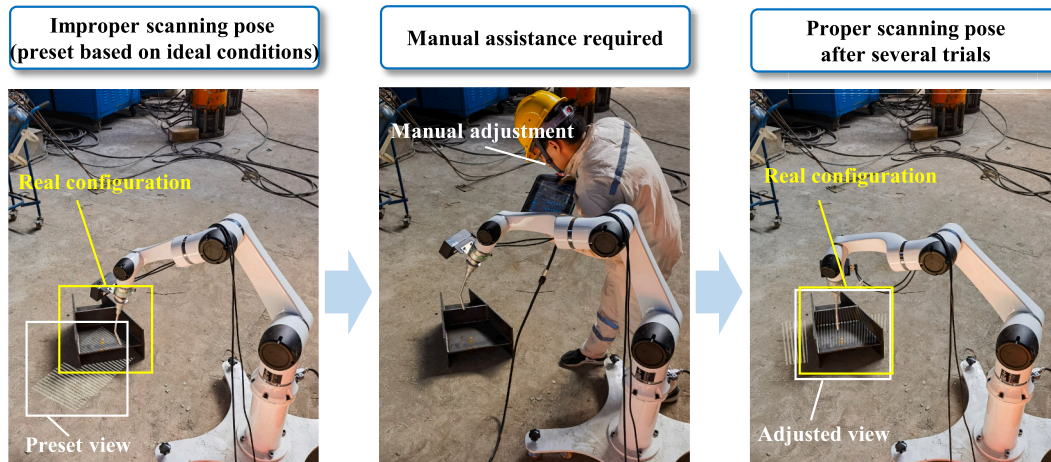


Fig. 1. Manual assistance is required in the current active vision-based seam detection framework.

only raises the cost but also limits the workpiece's size. Loukas et al. [18] employed a QR code tag to assist in locating the workpiece, but the process of affixing the QR code tag to the workpiece itself is labor-intensive.

To achieve automatic, adaptive, and economic seam detection in real production, a self-adaptive seam detection framework leveraging pose estimation of the target workpiece has been proposed in this paper. Pose estimation driven by deep neural networks allows the calculation of rotation and translation parameters in 3D Euclidean space using a standard RGB camera [19–22]. This non-contact measurement method not only inherits the cost advantages of visual sensing but also possesses the capability to adapt to dynamic, unconstructed welding environments. Intelligent and cost-effective seam detection is achieved by directly establishing the pose of the target 3D workpiece from an overall workspace 2D image, which enables automatic recognition of uncertainties in workpiece configurations and variances in working environments. A fully autonomous seam detection process is realized by strategically utilizing the estimated workpiece pose as guidance for the subsequent robot navigation and active vision measurement. It is pointed out that the robotic welding system makes decisions independently throughout the seam detection process, which means no human assistance is needed in the robotic welding process, including the workpiece pose estimation, robot navigation, structured light sensor measurement, seam extraction, and welding trajectory planning.

The rest of the paper is organized as follows. Section 2 outlines the overall seam detection framework. In Section 3, the enhanced algorithm for workpiece pose estimation is described, along with a rendering-based strategy for synthetic training dataset generation. Techniques related to automatic robot alignment using the estimated workpiece pose, point cloud processing, target seam extraction, and welding trajectory generation are presented in Section 4. Section 5 provides experimental results. Major conclusions, shortcomings, and potential future work are highlighted in Section 6.

2. Seam detection framework

A self-adaptive seam detection framework that leverages a reliable estimation of the workpiece pose from the overall workspace is proposed

in this research. As shown in Fig. 2, an autonomous robotic welding process is achieved utilizing the proposed seam detection framework through the following steps: (i) workpiece pose estimation, (ii) automatic robot navigation and structured light sensor scanning (point cloud acquisition), (iii) seam extraction and welding trajectory planning (point cloud processing), and (iv) robotic welding execution.

The proposed framework utilizes a workpiece pose estimation-based strategy to enhance both the adaptability and automation of weld seam detection. Through an enhanced workpiece pose estimation algorithm, the framework generates a reliable pose estimation of the target workpiece from an overall workspace image captured by the passive vision sensor. The estimated workpiece pose contains sufficient 3D information and enables the robotic welding system to accurately interpret the actual configuration of the target workpiece. As shown in Fig. 3, based on this real-time configuration, the system can autonomously determine the necessary robot movements to achieve an optimal pose for subsequent active vision sensor measurements. The introduction of pose estimation allows the robot to autonomously optimize the structured light sensor scanning pose in a cost-efficient manner, eliminating the need for human intervention while dramatically improving seam detection efficiency. This study strategically combines the wide field of view offered by the passive vision sensor and the high precision of the active vision sensor, optimizing both coverage and accuracy of the seam detection framework.

3. Workpiece pose estimation

This section develops an enhanced pose estimation algorithm that enables the prediction of reliable estimation of the target workpiece pose from a broad workspace image. The following subsections detail the architecture of the enhanced pose estimation algorithm, the rendering-based synthetic data generation strategy, and the creation of a diverse test dataset.

3.1. An enhanced two-stage pose estimation algorithm

A two-stage strategy is constructed to predict a reliable pose of the target workpiece. In the first stage, the algorithm establishes the 2D–3D

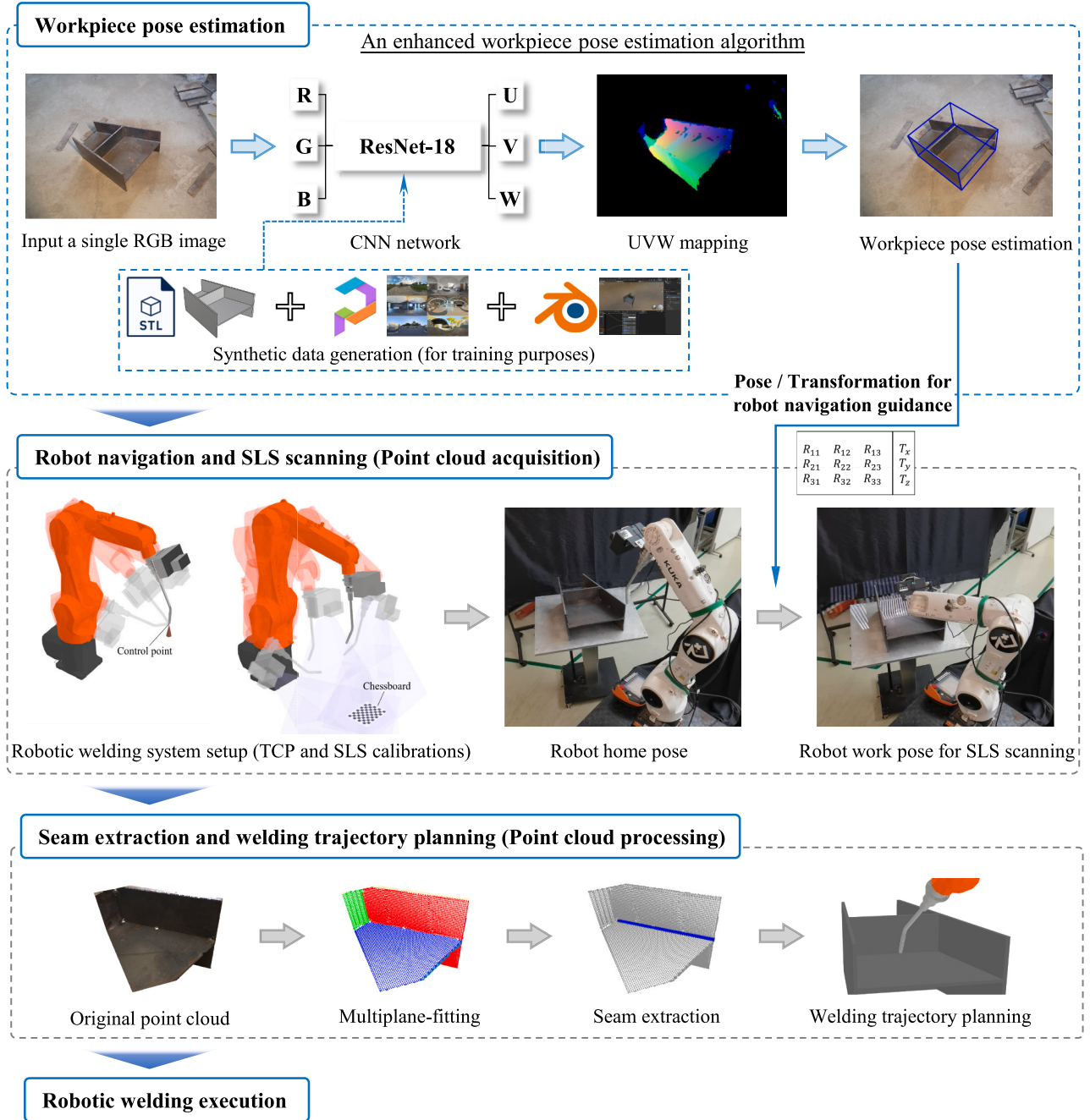


Fig. 2. Proposed seam detection framework leveraging workpiece pose estimation.

correspondence by mapping each pixel in the image captured by the RGB camera (i.e., 2D space) to a UVW coordinate system [23] on the target workpiece model (i.e., 3D space). In the second stage, the algorithm generates a reliable pose of the target workpiece using an EPnP-RANSAC flow. The architecture of the pose estimation algorithm is shown in Fig. 4.

Compared with the end-to-end method, this study constructs dense features in the form of UVW as an intermediate state to establish a

2D—3D mapping, thereby avoiding the need for direct estimation of the highly non-linear mapping from the 2D-pixel space to 3D geometrical quantities [24]. Serving as a dense correspondence framework, UVW mapping provides higher robustness against mismatched points. The pose is further calculated using the EPnP-RANSAC algorithm according to the 2D—3D mapping relationship. The algorithm employs a stochastic, iterative strategy for pose estimation, which not only expedites the computational process but also mitigates cumulative error

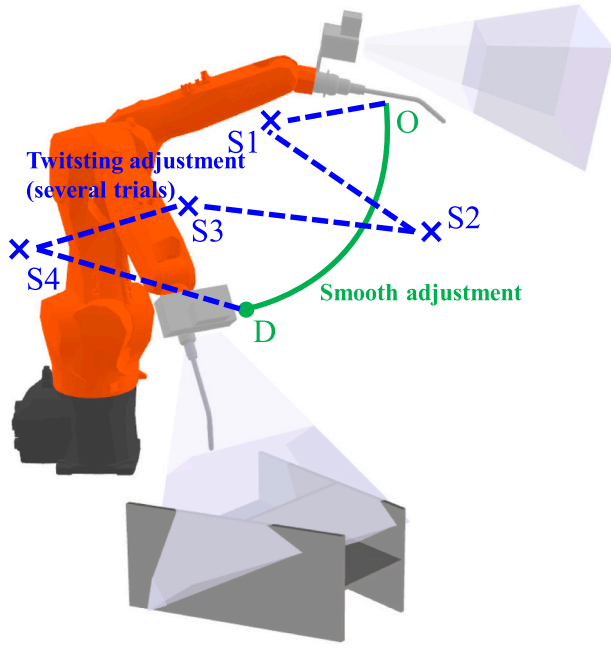


Fig. 3. Efficiency of the proposed seam detection framework.

propagation typically observed in purely iterative methods. This comprehensive method enhances both the reliability and efficiency of pose estimation. The methodology is outlined by the pseudo-code Algorithm 1.

Algorithm 1 Two-Stage 6D Object Pose Estimation

Require: Input single RGB image I

Require: Pre-trained deep network $\mathcal{F}_\theta$

Require: Camera intrinsic parameters $\mathbf{K}$

Require: 3D object model $\mathcal{M}$

Stage 1: UVW Map Prediction via Deep Learning

1. Predict per-pixel normalized 3D coordinates:

$$\mathbf{U}, \mathbf{V}, \mathbf{W} = \mathcal{F}_\theta(I)$$

Stage 2: 6D Pose Estimation via EPnP-RANSAC

1. Extract valid correspondences:

$$\mathcal{P} = \{(u_i, v_i, \mathbf{U}_i, \mathbf{V}_i, \mathbf{W}_i)\}$$

2. Estimate pose $(\mathbf{R}, \mathbf{t})$ using EPnP-RANSAC:

$$(\mathbf{R}, \mathbf{t}) = \text{EPnP-RANSAC}(\mathcal{P}, \mathbf{K}, \mathcal{M})$$

Output: Estimated 6D pose $(\mathbf{R}, \mathbf{t})$

3.1.1. Establishment of 2D—3D correspondence (1st stage)

An encoder-decoder architecture was constructed to predict 2D—3D correspondence using a single RGB image as input. As shown in Fig. 4, the encoder uses the ResNet-18 model, while the decoder comprises three branches, each corresponding to the U, V, and W channels, respectively. The network was trained to generate a dense UVW map corresponding to the input image. The composite loss function L for network parameters optimization is defined as:

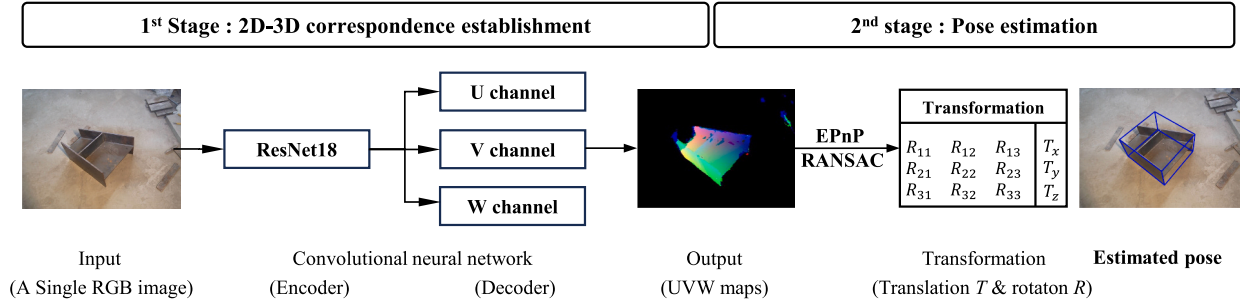


Fig. 4. Enhanced two-stage pose estimation algorithm.

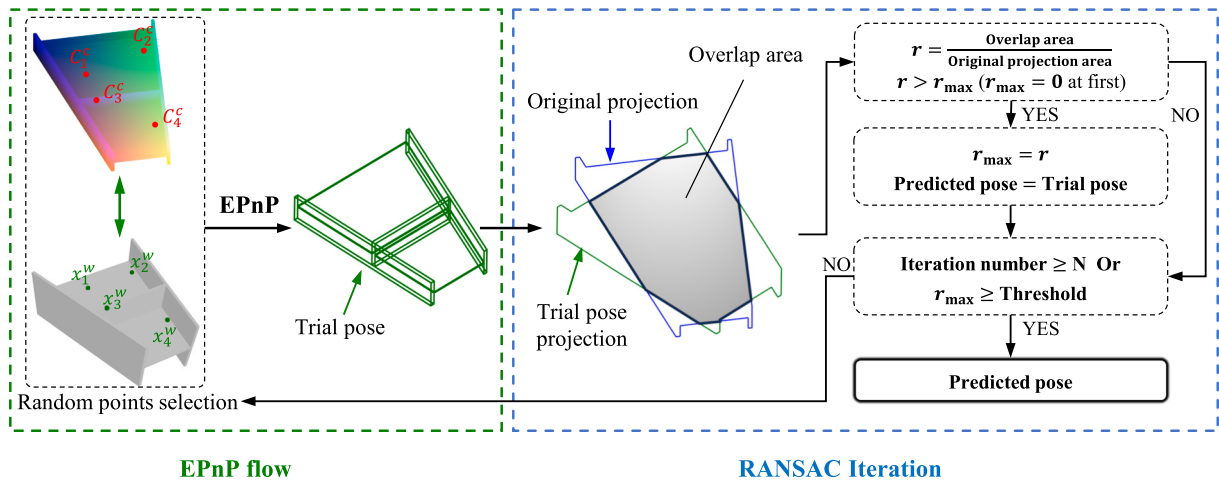
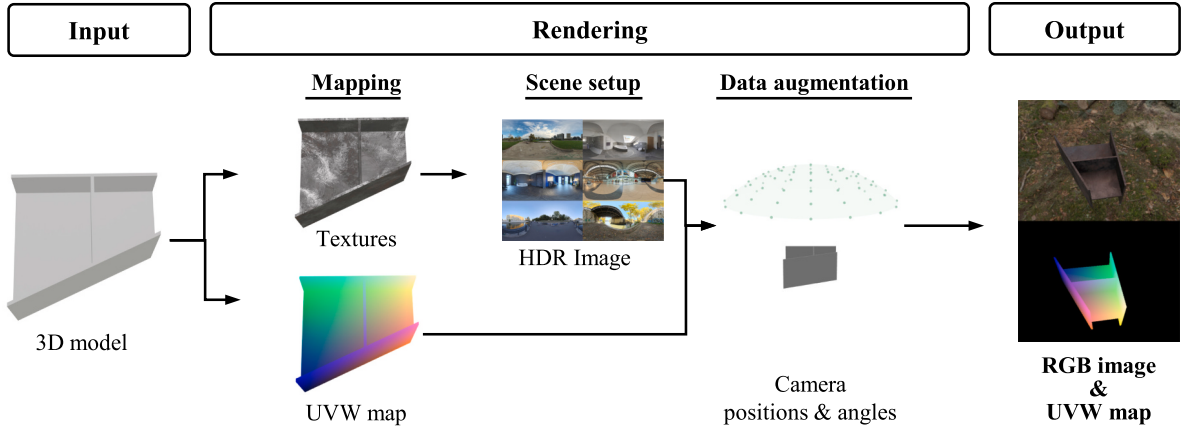
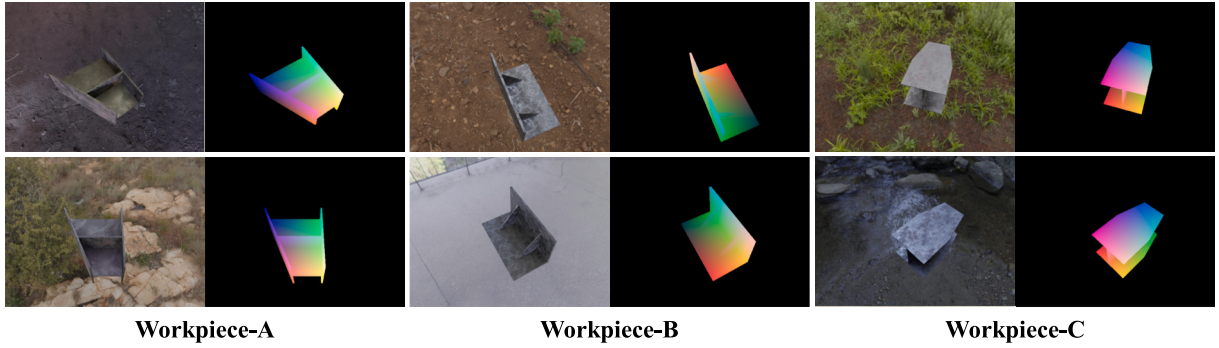


Fig. 5. EPnP-RANSAC workflow.



(a) Blender-based strategy for synthetic training data generation



(b) Representatives of diverse training data

Fig. 6. Training dataset creation.

$$L = L_U + L_V + L_W$$

(1)

where L_U , L_V , and L_W are the losses corresponding to the U, V, and W channels, respectively. Each of these losses is modeled using a multi-class cross-entropy function.

3.1.2. Workpiece pose estimation (2nd stage)

In the 2nd stage of the algorithm, an efficient workflow is established to generate an accurate transformation (i.e., pose) for the alignment of the 3D workpiece model with the observed 2D object in the image. The workflow highlights a stochastic, iterative approach which filters out incorrectly predicted 2D—3D correspondence and an enhancement to the computational efficiency which effectively handles large-scale point sets.

As shown in the workflow in Fig. 5, four random 2D—3D correspondences were selected in each iteration for the EPnP algorithm [25] to generate a trial pose. An evaluation of the trial pose is subsequently conducted by calculating the overlap ratio (r) between the projection of the trial pose and the projection of the corresponding UVW map. RANSAC (Random Sample Consensus) [26] is employed in the iteration process to ensure an accurate estimation of the workpiece pose. In this study, RANSAC works by repeatedly comparing different trial poses generated by the EPnP algorithm and keeping track of the trial pose that fits the best (i.e., the trial pose with the highest overlap ratio). The EPnP-RANSAC workflow is terminated either when an acceptance threshold of $r > 0.99$ is achieved or when the maximum number of iterations (up to 1000 in this study) is reached. The best-fit pose determined by the EPnP-RANSAC workflow is taken as the output workpiece pose for the entire pose estimation algorithm.

3.2. Dataset for pose estimation algorithm

High-quality datasets are essential for robust pose estimation. This study proposes a Blender-based synthetic data generation method to ensure the diversity and scalability of the training set while constructing a real-world test set to provide a reliable benchmark for performance evaluation.

3.2.1. Synthetic training dataset

Creating real-world training data for steel workpieces is challenging as their highly reflective surfaces always capture unwanted environmental reflections. Variations in lighting conditions and surface textures could also drastically alter the appearance of the workpieces. These factors make real-world data creation costly and complex [27], highlighting the need for alternative methods like synthetic data generation [28].

Conventional pose estimation algorithms commonly dealt with texture-rich and low-reflective objects with the associated training data generated using illumination-based approaches [23]. However, a trial study revealed that the traditional illumination-based approaches [29] performed inadequately in generating training data for pose estimation algorithms involving steel workpieces, primarily due to the reflective nature of steel surfaces. Thus, this study developed a rendering-based synthetic data generation strategy to simulate realistic surface textures of steel workpieces for comprehensive training purposes (see Fig. 6(a)).

In general, each synthetic item in the training dataset consists of a realistic image paired with its corresponding UVW map. The image creation process starts with a detailed 3D model of the target workpiece. Scenes are set up by placing the 3D model in various configurations, applying textures for steel workpieces, and incorporating different backgrounds and lighting conditions utilizing available High Dynamic

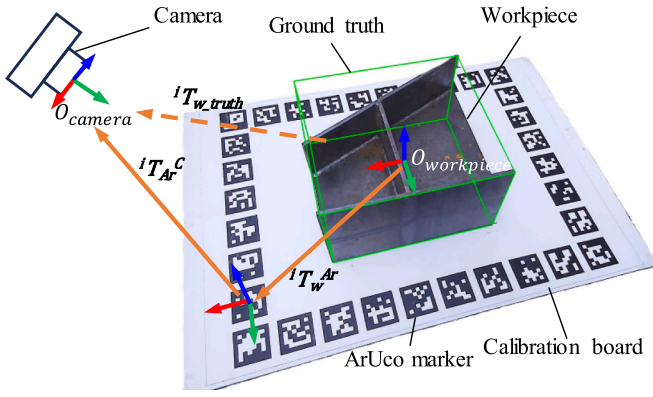
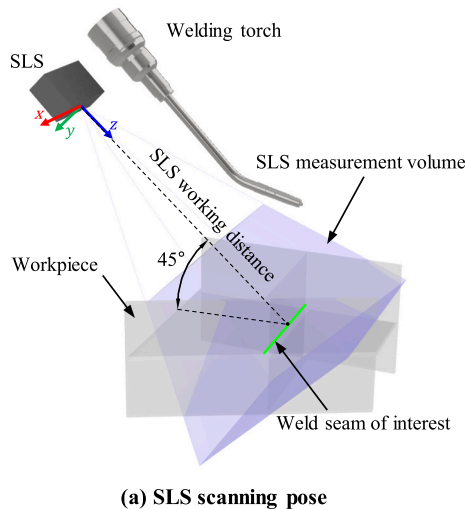


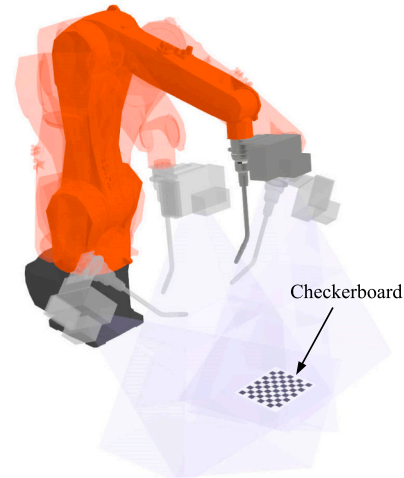
Fig. 7. Ground truth pose determination in test data.

Range (HDR) images [30] sourced from a public image library [31]. A virtual camera is then positioned within the scene to capture images from diverse viewpoints and angles. The camera positions are strategically distributed on a Fibonacci lattice across a hemisphere above the workpiece to simulate various camera poses. A Blender-based pipeline [32] is exploited in this study for realistic rendering. The UVW map is generated utilizing the same 3D model employed in the image creation process. It could be noticed that the color channels of the background in the UVW map were set to zero (i.e., displayed as black) to implicitly identified the background. Leveraging Blender's advanced rendering capabilities, this strategy faithfully replicates the textural properties and high-reflectivity characteristics inherent to steel structures. Taking into account the accessibility of models within the steel structure field, the proposed synthetic data generation strategy facilitates the efficient and scalable creation of training datasets tailored to various workpiece types. By significantly reducing the costs associated with data collection, this framework offers a robust and actionable data foundation to support the pose estimation-based seam detection algorithm proposed in this research.

Overall, the generated training dataset comprises of a diverse set of examples with various geometries, textures, backgrounds, camera poses, and light conditions explicitly considered. The dataset is sufficiently large enough to ensure comprehensive coverage and enhance the learning capacity of the developed pose estimation algorithm, with a total of 93,657 items included in the dataset. A group of representative diverse training data is shown in Fig. 6(b).



(a) SLS scanning pose



(b) SLS calibration

Fig. 8. Structured light sensor setup.

3.2.2. Test dataset

This section creates a high-quality test dataset to evaluate the performance of the enhanced two-stage pose estimation algorithm. Each entry in the test dataset consists of an image of the target workpiece captured from real-world scenes, paired with the ground truth pose of the workpiece. To create diverse images, variances in backgrounds (such as asphalt pavement, concrete slab, and epoxy floor), workpiece configurations, lighting conditions (natural light and artificial light), and camera viewpoints and angles, are explicitly considered. In total, 1226 real-world images were captured for this comprehensive test dataset.

To accurately determine the ground truth pose of the observed workpiece in the test images, a calibration board was developed with 32 ArUco markers [33] uniformly distributed along its perimeter (see Fig. 7). The markers are arranged in such a manner to ensure a considerable number of markers are always visible regardless of camera viewpoints or angles. Provided that a total of N markers ($i = 1$ to N) are identified in an image, the calculated pose of the workpiece relative to the camera based on the i^{th} identified ArUco marker (${}^i T_{w,truth}$) is established by:

$${}^i T_{w,truth} = {}^i T_{Ar}^C {}^i T_w^{Ar} \quad (2)$$

where ${}^i T_{Ar}^C$ is the pose of the i^{th} ArUco marker within the camera coordinate system, determined using available built-in functions from the OpenCV library [34] and taking into account the camera intrinsics and the physical size of the marker. Whereas, ${}^i T_w^{Ar}$ represents the workpiece pose in the coordinate system of the i^{th} identified marker.

Significantly deviated values are detected and removed from the calculated pose collection using the Isolation Forest algorithm [35]. The ground truth pose is then taken as an average of the rest calculated workpiece poses using Eq. (3). The Singular Value Decomposition (SVD) method is applied to the covariance matrix formed by quaternions, providing a robust way to compute the mean rotation from a set of rotations [36].

$$T_{w,truth} = \text{Avg} ({}^i T_{w,truth}) \quad (3)$$

4. Weld seam detection

The proposed seam detection framework leverages the reliable estimation of the target workpiece pose as guidance and enables automatic robot navigation for subsequent structured light sensor (SLS) measurements, and ultimately achieves a fully autonomous seam detection for robotic welding. The following subsections describe the methodologies and specific details corresponding to robot navigation, SLS scanning

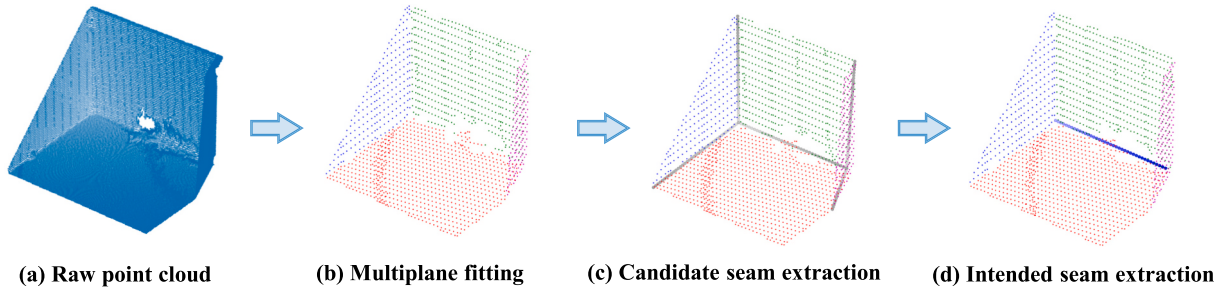


Fig. 9. Seam extraction.

(point cloud acquisition), weld seam extraction (point cloud processing), and welding trajectory planning.

4.1. Automatic robot navigation and SLS scanning

This section presents the preparations for the robotic system under consideration prior to the real-world application of the proposed pose-estimation framework. The following subsections detail the SLS setup configuration and robot navigation strategy.

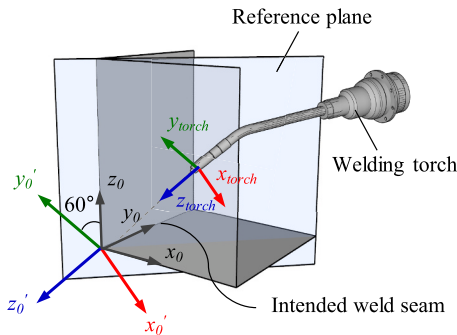
4.1.1. SLS setup

This study uses an SLS to generate a high-precision point cloud of the target workpiece and the intended weld seams. In order to ensure a complete and accurate data capture, an optimal scanning pose was pre-defined for the SLS within the workpiece coordinate system. This scanning pose allows the SLS to fully cover the weld seam within its field of view while working from an ideal distance away from the weld seam of interest. The actual dimensions of other objects in the environments, including the robot, welding torch, camera, and the target workpiece, were considered to avoid occlusions. A schematic of the SLS scanning pose is illustrated in Fig. 8(a). The calibration of the SLS using a checkerboard [37] to establish the transformation between the SLS and the robot is shown in Fig. 8(b).

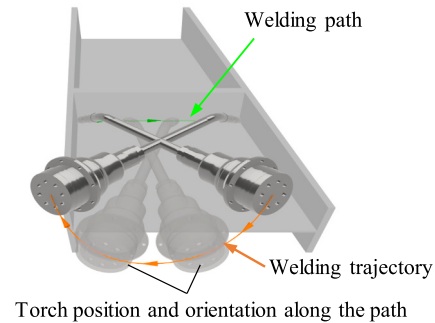
4.1.2. Robot motion for subsequent SLS scanning

The transformation of the robot from the home position to an optimal working pose is automatically established by the system via consideration of the predicted workpiece pose (T_w^R) using Eq. (4). In the equation, T_w^C is the transformation corresponding to the estimated workpiece pose. Whereas, T_C^R is a transform matrix which converts the workpiece pose from the camera view to the robot view, which is derived from the RGB camera calibration process.

$$T_w^R = T_C^R \cdot T_w^C \quad (4)$$



(a) Torch orientation at start point



(b) Welding trajectory generation

Fig. 10. Welding trajectory planning.

It is mentioned that the robot control system is designed to manage the position and orientation of the tool control point (i.e., the torch tip) to execute the welding. The motion planning algorithm RRT* [38] is utilized to generate optimal motions for the welding robot under consideration.

4.2. Point cloud processing and weld seam extraction

Noisy points in the point cloud acquired by the structured light sensor were removed prior to seam extraction. Points located below the welding platform and above the torch tip were eliminated to effectively reduce the dataset size. The point cloud was further filtered using a distance-based clustering method, where points with a specified distance from one another were grouped into clusters. The largest cluster was taken as the point cloud representing the target workpiece.

RANSAC algorithm was utilized to detect multiple planes from the filtered workpiece point cloud through an iterative process (see Fig. 9 (b)). Weld seams were extracted as the intersection lines between the detected planes. It is mentioned that several candidate weld seams might be extracted using the approach described above (see Fig. 9(c)). To identify the intended seam for subsequent welding, all extracted seams were compared with the corresponding 3D workpiece model in which the target seam had been pre-labeled. As illustrated in Fig. 9(d), the extracted seam that exhibited the least variance from the labeled seam in the model was taken as the target seam in the point cloud. It is noted that all the seams considered in this research were unwelded.

4.3. Welding trajectory planning

This section establishes the welding trajectory, including the orientation and path of the torch, according to the extracted weld seam features. Considering that this study focuses on straight weld seams, the torch path is consistently aligned with the target seam, and a satisfactory welding trajectory is strategically generated by (i) determining the

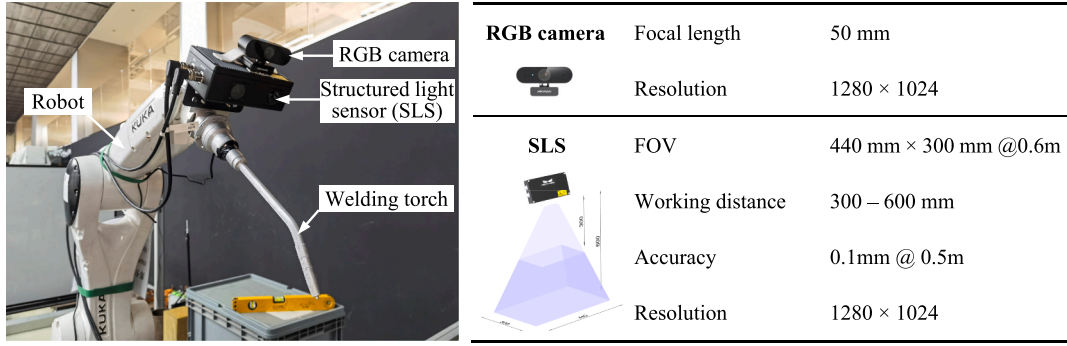
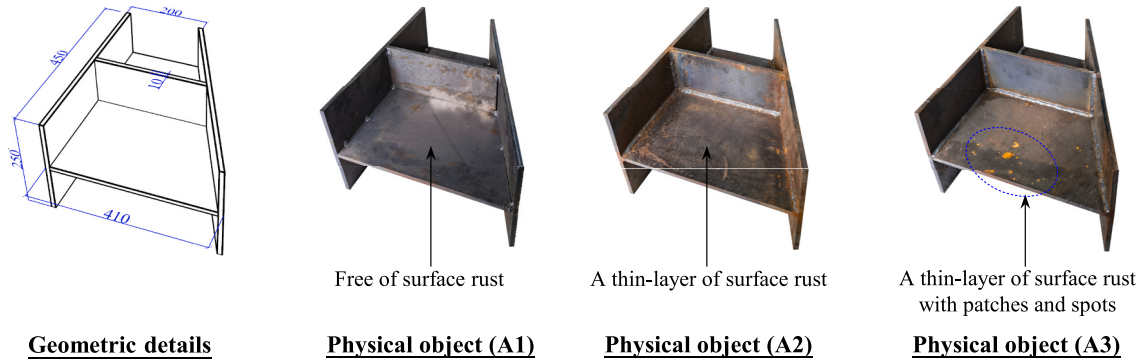
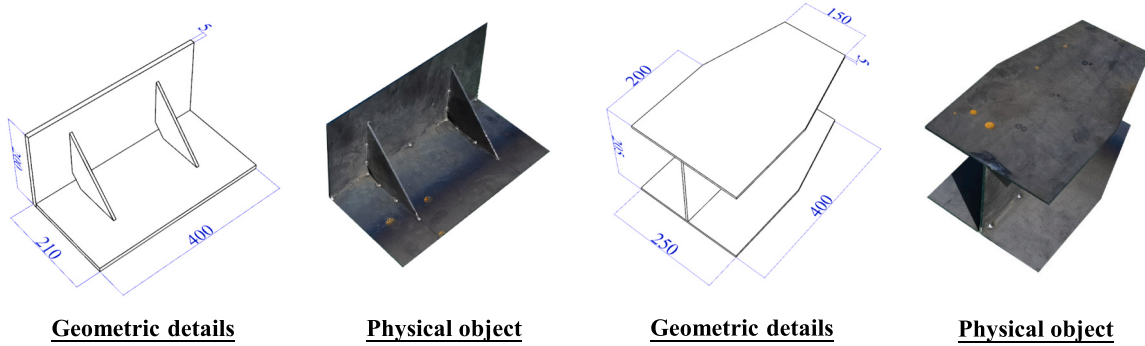


Fig. 11. Robotic welding system.



(a) Workpiece-A



(b) Workpiece-B

(b) Workpiece-C

Fig. 12. Geometric details and the surface textures of the experimental workpieces.

orientation of the welding torch at the start and end points of the intended seam (see Fig. 10(a)), and then (ii) interpolating the position and orientation of the torch at waypoints along the seam, ensuring a consistent linear motion along the weld path (see Fig. 10(b)).

Fig. 10(a) illustrates the determination of the welding torch orientation at the start point of an intended seam. A x_0 - y_0 - z_0 coordinate system is established at the start point with the y_0 -axis aligned along the intended seam and the z_0 -axis directed vertically upwards. A reference plane is then defined using the z_0 -axis and the angular bisector between the x_0 and y_0 axes. The x_0 - y_0 - z_0 coordinate system is rotated so that the new y_0' and z_0' axes lie on this reference plane, forming a new x_0' - y_0' - z_0' coordinate system. The orientation of the welding torch is set by aligning the torch coordinate system (x_{torch} - y_{torch} - z_{torch}) with the x_0' - y_0' - z_0' coordinate system, as shown in Fig. 10(a). In this study, the angle between the y_0' -axis and the z_0' -axis is set to 60°, such that the welding torch forms an idealized work angle relative to the intended seam. Though not

explicitly shown in the figure, the welding torch orientation at the endpoint of the intended seam is determined using the same approach. The tip of the welding torch is maintained at an appropriate distance (15 mm in this study) from the weld seam to prevent collisions between the welding torch and the workpiece.

5. Experiments and discussion

A series of experiments was conducted to validate the proposed framework. The subsequent subsections detail the experimental implementation, the evaluation of the enhanced pose estimation algorithm, and the performance assessment of the seam detection framework, along with analysis and discussion of the underlying causes of the results.

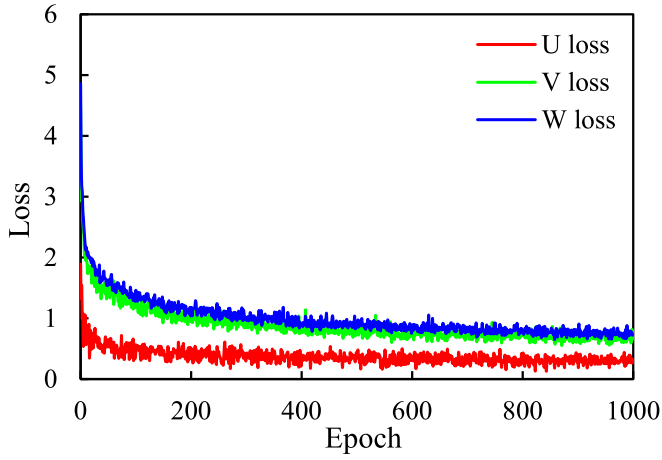


Fig. 13. Training losses.

5.1. Experimental implementation

This study conducted experiments on actual robotic system and real workpieces, with detailed specifics described in the following subsections.

5.1.1. Welding robot system

As illustrated in Fig. 11, the self-adaptive robotic welding system primarily consisted of a welding robot, an RGB camera (a passive vision sensor), a structural light sensor (an active vision sensor), and an industrial computer. Both the RGB camera and the SLS were mounted on the robot. The key statistics of these sensors are also shown in Fig. 11.

5.1.2. Experimental workpieces

This study evaluated the performance of the proposed seam detection framework using three types of experimental workpieces with different geometric dimensions (see Fig. 12). Specifically, there are three specimens of Workpiece-A, each featuring a distinct surface texture. Workpiece-A1 was mostly free of rust, while Workpiece-A2 had a thin uniform layer of rust. In contrast, Workpiece-A3 exhibited both general rust and prominent localized rust patches and spots. These workpieces under consideration were satisfactory representations of the steel members and connections commonly used in the construction industry, particularly in terms of welding.

5.1.3. Training of the enhanced workpiece pose estimation algorithm

The enhanced workpiece pose estimation algorithm was trained on a personal computer with an AMD EPYC 7452 processor (32 total cores) and an NVIDIA GeForce RTX 3090 GPU (24GB of memory), utilizing the PyTorch framework on a Linux system. The Adam optimizer was

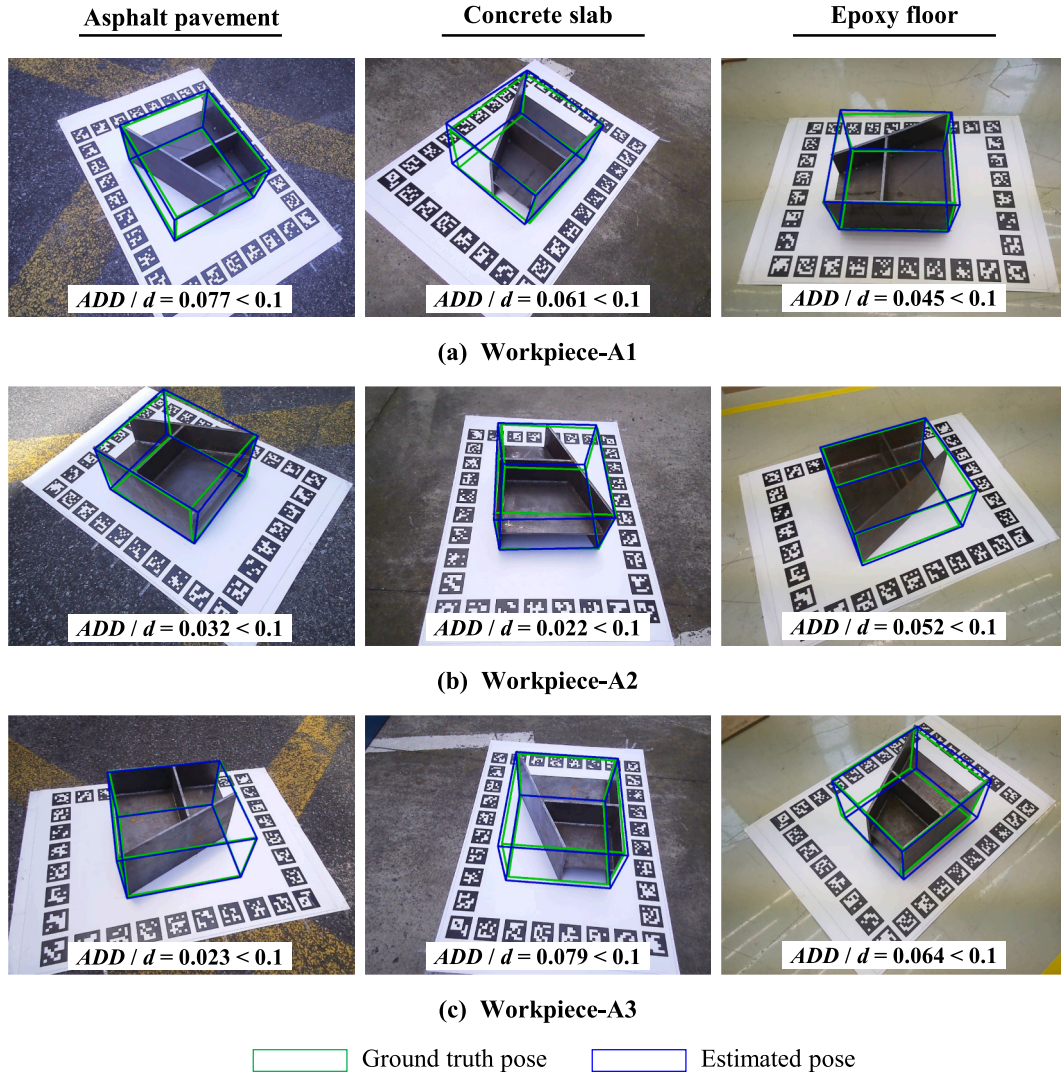


Fig. 14. Accuracy evaluation of the enhanced two-stage pose estimation algorithm for workpieces with distinct surface textures under various backgrounds.

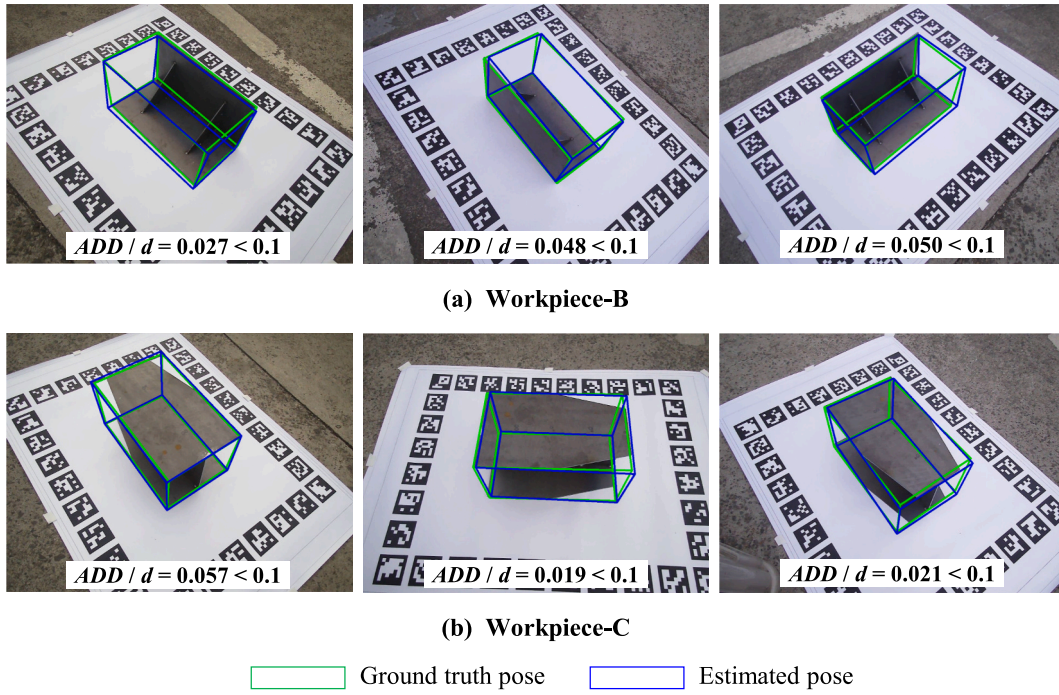


Fig. 15. Accuracy evaluation of the enhanced two-stage pose estimation algorithm for workpieces with diverse geometries.

employed with a learning rate of 0.0003 and a weight decay of 0.00004. The training process consisted of 1000 epochs, with each epoch consisting of 80 batches of size 14. The training losses associated with the U channel, V channel, and W channel are shown in Fig. 13. It can be observed that the training loss converged way before the training process was completed.

5.2. Performance evaluation of enhanced pose estimation algorithm

This section evaluates the performance of the enhanced two-stage pose estimation algorithm, the key component of the overall seam detection framework. A comprehensive test dataset consisting of 1226 real-world images of the experimental workpieces was used to validate the enhanced algorithm. The ground truth pose for each workpiece in the images was determined using the method outlined in Section 3.2.2, while the estimated pose of the corresponding workpiece was predicted by the enhanced algorithm. In order to reflect the characteristics of variable textures and unstructured environment in steel workpiece welding operations, a dataset was constructed featuring workpieces with various geometric dimensions, distinct surface textures placed in varied illumination conditions and against different backgrounds (i.e., asphalt pavement, concrete slab, and epoxy floor). In each scene, images were taken from different angles. Fig. 14 and Fig. 15 show a selection of representative images from the test dataset. In each image, both the ground truth pose (highlighted by green boxes) and the estimated pose (highlighted by blue boxes) are clearly identified. It can be observed that the enhanced pose estimation algorithm provided accurate predictions with only minor differences between the ground truth and the estimated pose.

Furthermore, the performance of the proposed pose estimation algorithm is evaluated using the Average Distance (ADD) metric as proposed in [39]. The effectiveness of such evaluation metrics in assessing pose estimation accuracy has been well acknowledged by researchers (see [19–22], and others). By converting the ground truth pose and the corresponding estimated pose of a particular workpiece into point sets M and M' , the average 3D distance (ADD) between all model points x in the ground truth pose and the estimated pose is established as:

Table 1

Performance evaluation of the enhanced pose estimation algorithm.

Experimental object	Background	Total number of test images	Observed workpiece pose correctly estimated	* ADD ¹⁰
Workpiece-A1	Asphalt pavement	131	113	86.26 %
	Concrete slab	112	100	89.29 %
	Epoxy floor	100	84	84.00 %
		343	297	86.59 %
Workpiece-A2	Asphalt pavement	102	87	85.29 %
	Concrete slab	96	82	85.42 %
	Epoxy floor	98	84	85.71 %
		296	253	85.47 %
Workpiece-A3	Asphalt pavement	130	115	88.46 %
	Concrete slab	110	88	80.00 %
	Epoxy floor	130	110	84.62 %
		370	313	84.59 %
Workpiece-B	Concrete slab	111	94	84.68 %
Workpiece-C	Concrete slab	106	91	85.85 %
Total		1226	1048	
Mean				85.42 %

* Percentage of the correctly estimated poses by the enhanced algorithm, where the ADD distance falls within the threshold $ADD < 0.1d$.

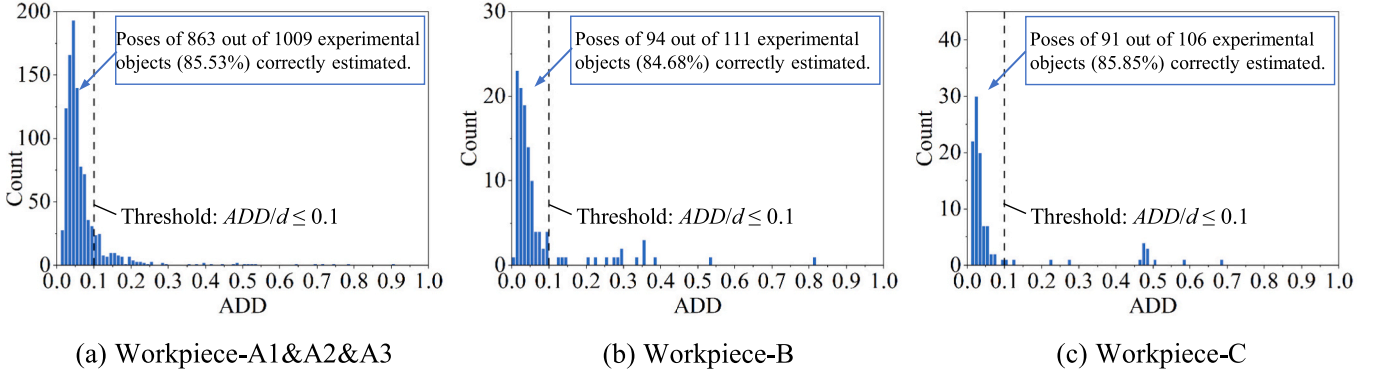


Fig. 16. Consistency of the enhanced pose estimation algorithm for diverse workpiece geometries.

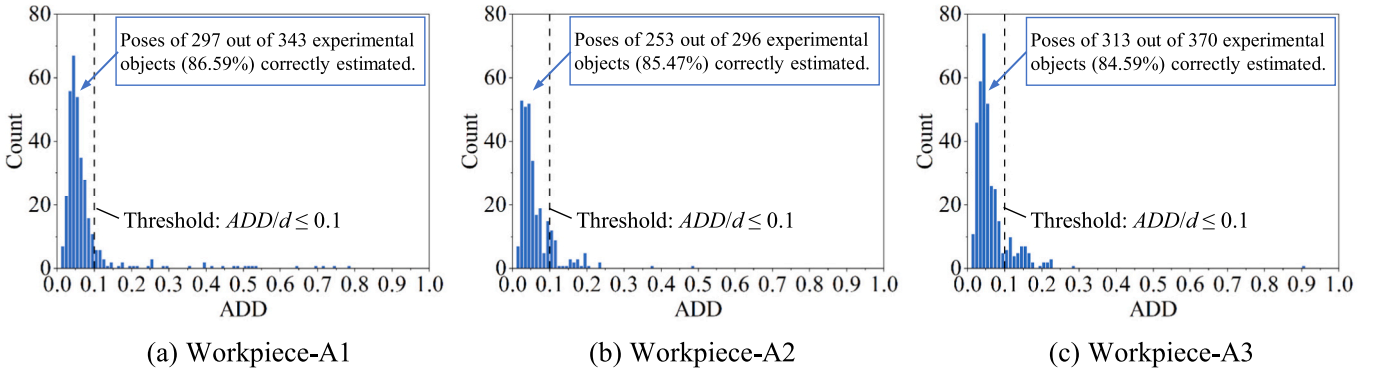


Fig. 17. Consistency of the enhanced pose estimation algorithm for various surface textures.

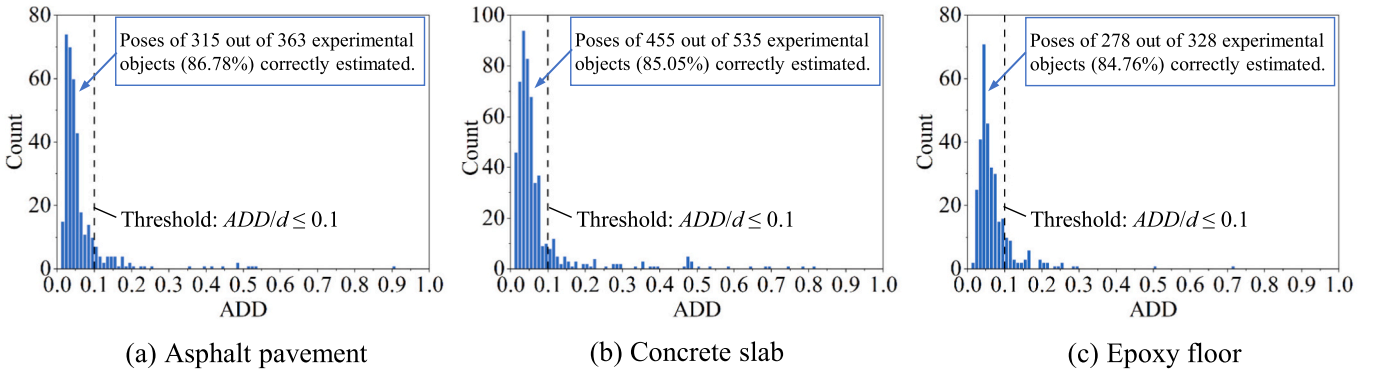


Fig. 18. Consistency of the enhanced pose estimation algorithm for various backgrounds.

$$ADD = \text{avg}_{x \in M} \|(\mathbf{R}\mathbf{x} + \mathbf{t}) - (\hat{\mathbf{R}}\mathbf{x} + \hat{\mathbf{t}})\| \quad (5)$$

where $\mathbf{R}$ and $\mathbf{t}$ represent the rotation and translation matrices of the ground truth pose, whereas, $\hat{\mathbf{R}}$ and $\hat{\mathbf{t}}$ denote the rotation and translation matrices of the estimated pose.

It is a widely accepted standard that if the ADD distance calculated using Eq. (5) is less than 10 % of the 3D model diameter (d), the estimated pose is considered correct and satisfactory. A complete analysis of the test datasets is provided in Table 1. Key statistics for 11 subsets, including the experimental workpiece under consideration, background, total number of images in each subset, number of images in which the observed workpiece pose were correctly estimated, and the percentage of correctly estimated workpiece poses by the enhanced algorithm (ADD^{10} , where the ADD distance falls within the threshold $ADD < 0.1d$),

are clearly outlined in the table.

As could be observed in the last column of Table 1, the enhanced algorithm provided a consistent prediction to the poses of target workpieces across all sub-test datasets, with a lowest ADD^{10} of 80.00 % and a highest ADD^{10} of 89.29 %. Totally, the ADD^{10} across the entire test dataset reached 85.42 %, with only minor variation among individual subsets. The consistent performance across diverse scenarios highlights the robustness and adaptability of the proposed pose estimation algorithm in effectively capturing and learning the intrinsic features of various steel workpieces.

Further evaluation of the algorithm's performance is conducted from three dimensions: geometry, surface texture, and background. Fig. 16 illustrates the performance of the enhanced pose estimation algorithm in terms of the ADD distance for diverse workpiece geometries. In all cases, a substantial number of target workpieces were accurately predicted,

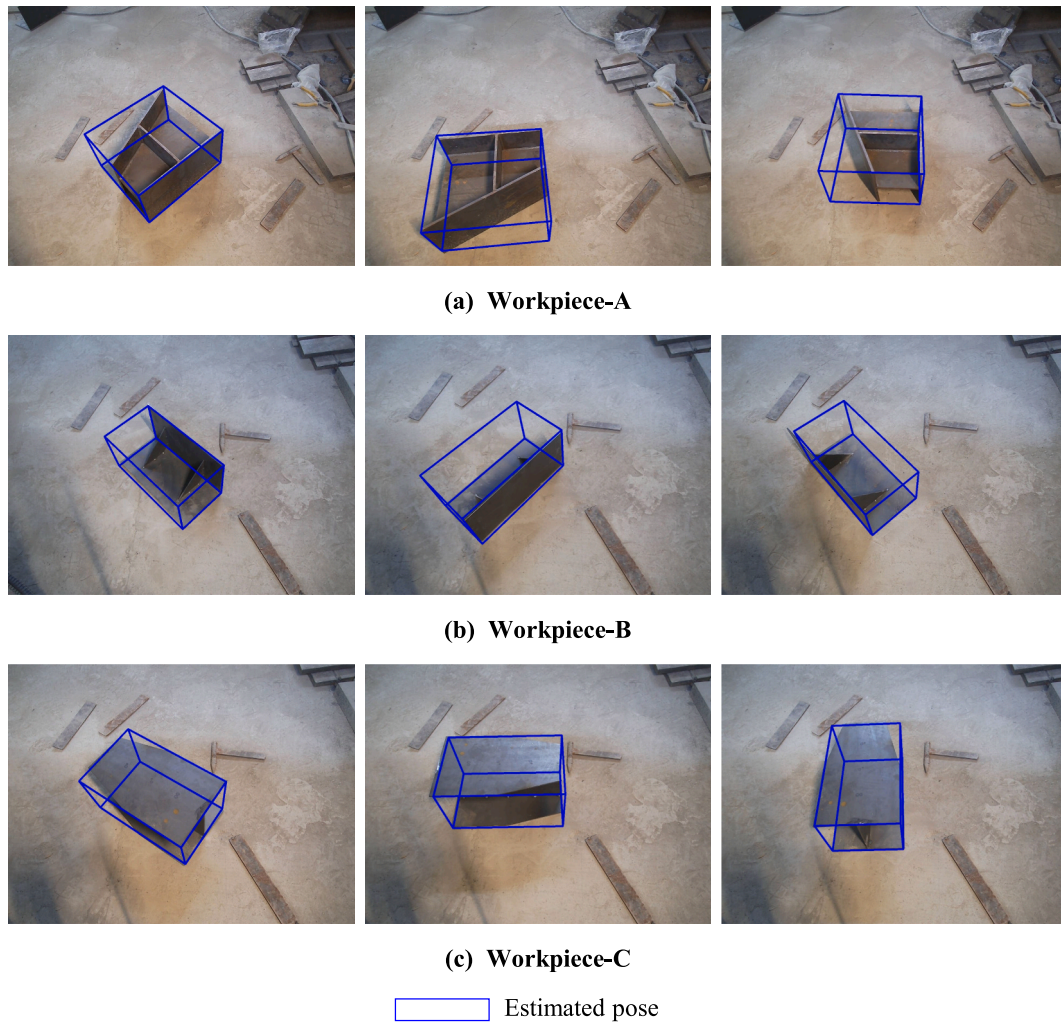


Fig. 19. Performance of the pose estimation algorithm in actual production.

meeting a threshold of $ADD < 0.1d$. Fig. 17 demonstrates the algorithm's performance across various surface textures. Fig. 18 shows the algorithm performance across different backgrounds. The results clearly indicate that the enhanced pose estimation algorithm exhibited consistent and reliable performance, providing accurate predictions for various objects in distinct backgrounds.

It is pointed out that the calibration board was used exclusively during the preparation of the test dataset. Neither the calibration process nor the calibration board is required for deploying the pose estimation algorithm in real-world applications. The performance of the pose estimation algorithm in practical and industrial scenarios is depicted in Fig. 19. It is evident that the proposed pose estimation algorithm achieves reliable and accurate predictions for diverse workpiece types and configurations in operational environments.

5.3. Performance evaluation of the proposed seam detection framework

This section provides a comprehensive assessment of the proposed seam detection framework, conducting a multidimensional analysis to rigorously evaluate its performance across accuracy, adaptability, and efficiency.

5.3.1. Accuracy of the proposed seam detection framework

Fig. 20 presents six representative seam detection results, including three vertically orientated seams (Seam-1, Seam-2, and Seam-3) and three horizontally orientated seams (Seam-4, Seam-5, and Seam-6). The

ground truth location of each target seam (highlighted in green) was established utilizing a teaching-playback method, while the predicted seam location (highlighted in blue) was generated through the proposed seam detection framework. It is clear from Fig. 20 that the predicted seam locations were consistent with their corresponding ground truth locations.

To further quantify the variance between the predicted seam and the ground truth seam, both seams were divided into 100 uniformly spaced points. The maximum error (ME, the largest distance between corresponding points) and the root mean square error (RMSE, the quadratic mean of the differences between the two point sets) were computed for each target seam. Consideration of all representative seams shown in Fig. 20, a minimum maximum error (ME) of 0.68 mm (Seam 2) and a minimum root mean square error (RMSE) of 0.15 mm (Seam 2) were observed. Whereas, the largest maximum error (ME) was only 1.56 mm (Seam 6) and the largest root mean square error (RMSE) was 0.48 mm (Seam 4). These results suggest that the variance between the predicted and ground truth seams was minimal.

5.3.2. Adaptability of the proposed seam detection framework

This section evaluates the adaptability of the proposed seam detection framework by conducting a series of comparative experiments on various workpieces in different environments. Fig. 21 showcases the process and the results of the comparative tests performed on two representative real-world working scenarios. Each test started with a consistent initial setup, where operators deliberately created random

Target seam location and orientation	Ground truth	Point cloud acquisition and processing	Seam detection	Performance evaluation
				<i>Error_ME</i> : 1.14 mm <i>Error_RMSE</i> : 0.37 mm Total time: 103 s
(a) Seam 1 (Vertical orientation)				
				<i>Error_ME</i> : 0.68 mm <i>Error_RMSE</i> : 0.15 mm Total time: 102 s
(b) Seam 2 (Vertical orientation)				
				<i>Error_ME</i> : 1.36 mm <i>Error_RMSE</i> : 0.34 mm Total time: 98 s
(c) Seam 3 (Vertical orientation)				
				<i>Error_ME</i> : 1.25 mm <i>Error_RMSE</i> : 0.48 mm Total time: 125 s
(d) Seam 4 (Horizontal orientation)				
				<i>Error_ME</i> : 0.82 mm <i>Error_RMSE</i> : 0.26 mm Total time: 115 s
(e) Seam 5 (Horizontal orientation)				
				<i>Error_ME</i> : 1.56 mm <i>Error_RMSE</i> : 0.27 mm Total time: 134 s
(f) Seam 6 (Horizontal orientation)				

Fig. 20. Performance of the proposed seam detection framework.

workpiece poses to simulate the uncertainty in workpiece configuration, aiming to replicate real-world scenarios and establish a realistic testbed for evaluating the proposed seam detection framework (see item ① in Fig. 21). These arrangements simulated common scenarios encountered in robotic welding industrial practices, in which the actual position and orientation of the target workpiece often deviate from the idealized configuration. The initial photographing pose of the robot is pre-set and fixed. By adopting this pose, the RGB camera can achieve satisfactory images. Meanwhile, a proper distance can be maintained between the end-effector (welding torch) and the workpiece, thus ensuring an adequate safety margin. A sampling frequency of 1.5 Hz was utilized for the SLS in the tests.

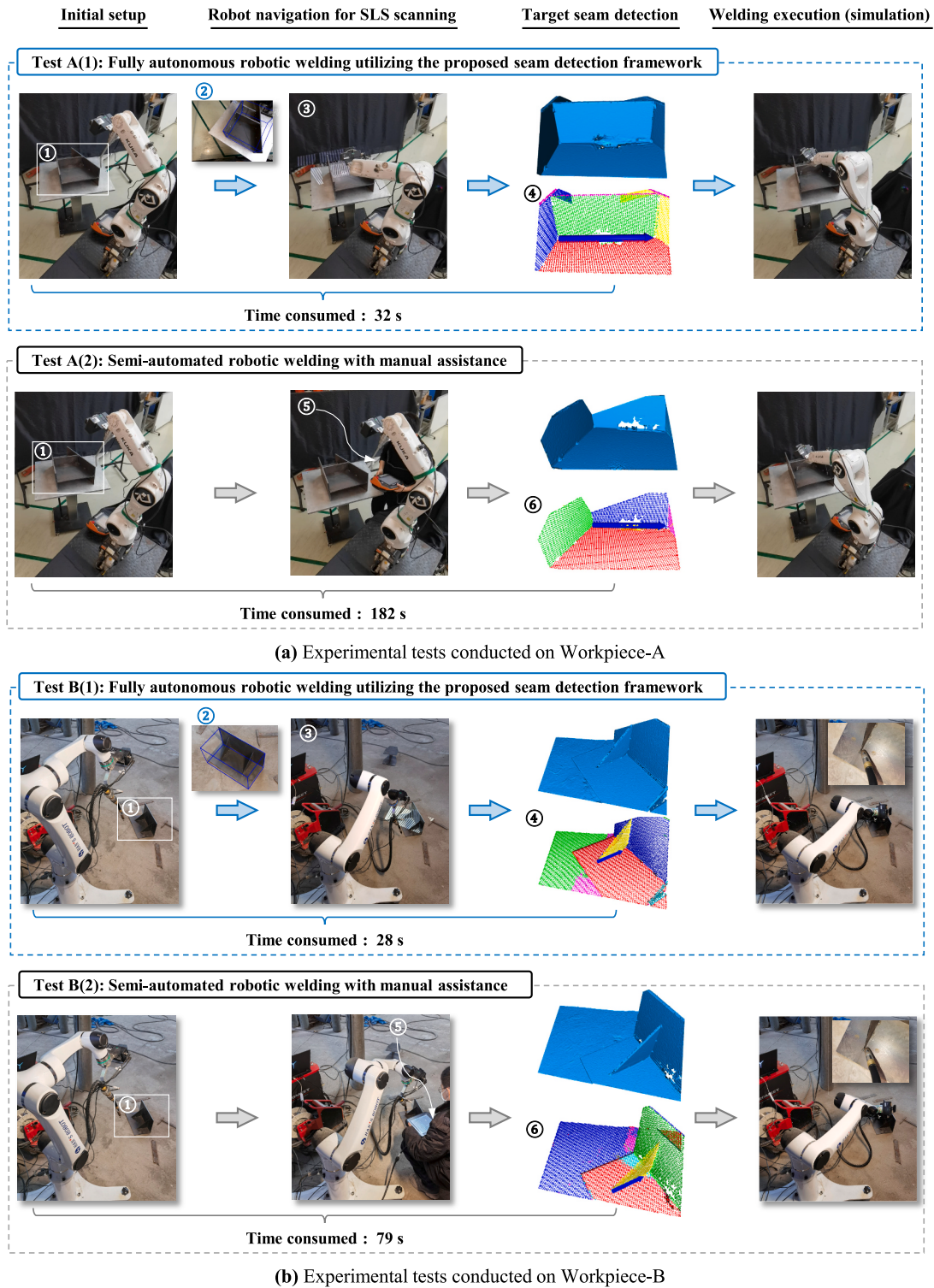
As shown in Fig. 21, in Test A(1) and Test B(1), a fully autonomous robotic welding process was achieved utilizing the proposed seam detection framework. The proposed framework enabled the robot to autonomously adjust for subsequent SLS scanning by leveraging the workpiece pose estimation strategy (see items ② and ③). The point cloud acquired by the SLS (see item ④) clearly shows that the proposed framework ensured the target seam was perfectly centered within the field of view of the SLS. This allowed the robot enhance both the quality and efficiency of the welding process that followed. It is noted that,

although actual welding operations were not performed during the experimental tests, the welding path of the end-effector (torch), as demonstrated in the attached video, closely aligned with the target seam, indicating satisfactory accuracy of the proposed framework.

Test A(2) and Test B(2) simulated the workflow typically adopted in industrial practices, where a semi-automated robotic welding process was performed. Assistance from an experienced operator was required to manually navigate the robot such that the target seam appeared in the field of view of the SLS (see item ⑤). In such a manual-assistance-required robotic welding process, however, the target seam was often not perfectly centered in the field of view of the SLS, as evident from the acquired point cloud data (see item ⑥). Such misalignment could potentially affect the accuracy of the seam detection and, consequently, the overall quality and efficiency of the welding process.

5.3.3. Efficiency of the proposed seam detection framework

Total 12 experiments were conducted on various types of workpieces. The time required for welding robots based on the proposed seam detection framework and previous seam detection framework [15] are summarized in Table 2. In traditional weld seam detection frameworks, the workpiece's pose cannot be pre-perceived. Consequently, manual



- ① The experimental workpiece randomly placed within the workspace
- ② Estimation of the workpiece pose by the proposed seam detection framework as guidance for subsequent robot navigation
- ③ An optimal SLS scanning position and orientation automatically achieved by utilizing the proposed seam detection framework
- ④ The target seam perfectly aligned at the center of the field of view of the SLS
- ⑤ An experienced operator required to manually navigate the robot for subsequent SLS scanning
- ⑥ The target seam fully within the field of view of the SLS but not centered

Fig. 21. Experimental tests.

Table 2

Efficiency evaluation of the robotic welding tasks: Proposed seam detection framework vs. Manual-assistance-required framework.

Experimental number	Time consumed (s)					
	SLS scanning & seam detection		Welding execution (simulation)		Total time duration	
	Proposed seam detection framework	Manual asst. Required [15]	Proposed seam detection framework	Manual asst. Required [15]	Proposed seam detection framework	Manual asst. Required [15]
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Test A-1 (Seam-1)	33 (↓80 %)	168	70	77	103 (↓58 %)	245
Test A-2 (Seam-2)	31 (↓82 %)	172	71	73	102 (↓58 %)	245
Test A-3 (Seam-3)	33 (↓59 %)	80	65	59	98 (↓29 %)	139
Test A-4 (Seam-4)	32 (↓82 %)	182	93	85	125 (↓53 %)	267
Test A-5 (Seam-5)	32 (↓81 %)	168	83	89	115 (↓55 %)	257
Test A-6 (Seam-6)	34 (↓31 %)	49	100	104	134 (↓12 %)	153
Test B-1	28 (↓65 %)	79	64	62	92 (↓35 %)	141
Test B-2	32 (↓70 %)	105	62	63	94 (↓44 %)	168
Test B-3	32 (↓58 %)	76	66	62	98 (↓29 %)	138
Test C-1	30 (↓76 %)	126	87	82	117 (↓44 %)	208
Test C-2	33 (↓70 %)	109	90	86	123 (↓37 %)	195
Test C-3	31 (↓51 %)	63	89	83	120 (↓18 %)	146
Average	32 (↓72 %)	115	78	77	110 (↓43 %)	192

pre-adjustment of the robotic arm is usually required to enable the structured light sensor (SLS) to accurately capture the point cloud of the target area. In contrast, the proposed framework enhances automation and efficiency by eliminating the need for manual intervention. For each test, the time spent on SLS scanning and seam detection, as well as the time taken for the robotic welding execution simulation, are explicitly provided.

Comparison between Columns (2) and (3) of Table 2 shows that, on average, the proposed seam detection framework reduced the SLS scanning and seam detection time by 72 %. Overall, the implementation of the proposed seam detection framework led to an average reduction of 43 % in the total duration of the entire robotic welding process, as shown in Table 2 Columns (6) and (7). These results suggest that the proposed seam detection framework significantly enhances the efficiency of the robotic welding process.

It can also be noticed from Table 2 Column (2) that, for each target seam under consideration, the time spent on SLS scanning and seam detection was consistent when utilizing the proposed seam detection framework. The consistency stems from the well-defined scanning pose of the robotic arm determined based on the estimated pose of the workpiece. As a result, the robotic arm follows a more straightforward trajectory between the initial pose and the working pose and therefore ensures stable and predictable time consumption. On the other hand, Column (3) of Table 2 highlights considerable variability in the time required for SLS scanning and seam detection when manual assistance was involved. This variability can be attributed to the operators' inability to intuitively perceive the scanning field of view of the SLS. An iterative process with repeated adjustments to the robotic arm is always required to achieve an acceptable scanning pose such that the SLS scanning range fully covers the target weld seam region.

The proposed seam detection framework strategically leverages the wide field of view of the passive vision sensor and the high precision of the active vision sensor, thereby improving coverage, accuracy, and efficiency in robotic welding seam detection. Comparative experiments demonstrate that the proposed framework consistently detects target seams with high accuracy and guides the robot to optimal working positions and orientations with high efficiency, without the need for a pre-defined workpiece setup or human supervision. A demonstration video of the comparative welding simulations can be viewed in the appendix.

6. Conclusions and future work

Both the adaptability and automation of the robotic welding system are improved by leveraging workpiece pose estimation. The real-time

pose of the workpiece is detected from an overall workspace image, automatically guides the robot to an optimal working position and orientation for subsequent structured light sensor measurements based on the detected workpiece pose, and therefore achieves a fully autonomous seam detection process without requiring pre-defined workpiece configurations or human intervention. Experimental results demonstrate that the framework significantly improves the efficiency of seam detection while exhibiting consistent performance across various target workpieces and diverse backgrounds.

An enhanced two-stage workpiece pose estimation algorithm is developed to generate reliable estimations of target workpiece poses from broad workspace images, employing a ResNet-18 convolutional neural network in the first stage to establish 2D–3D correspondences and produce a UVW map of the observed workpiece from the input image, while the second stage utilizes an EPnP-RANSAC workflow for accurate pose estimation. Additionally, a rendering-based synthetic data generation strategy is introduced to create realistic simulations of surface textures for steel workpieces, including their visual interactions with light and environmental elements, thereby ensuring high-quality training datasets that enhance the effectiveness and reliability of the proposed pose estimation algorithm.

By integrating pose estimation, the proposed seam detection framework effectively mitigates the uncertainties associated with workpiece configuration, thereby facilitating efficient and accurate welding operations in unstructured environments for structural steel welding. No manual assistance is needed in the robotic welding process, including the workpiece pose estimation, robot navigation, structured light sensor measurement, seam extraction, and welding trajectory planning. The proposed seam detection framework will significantly reduce the operational costs of welding robots and enhance production efficiency.

The main contributions of this paper include: (i) proposing a self-adaptive seam detection framework that strategically utilizes the estimated pose of the target workpiece to enhance the adaptability and automation of the seam detection process. It is pointed out that the robotic welding system makes decisions independently throughout the seam detection process. Compared to the traditional framework, no manual assistance is needed in the proposed framework; (ii) developing an enhanced workpiece pose estimation algorithm to robustly generate a reliable estimation of the target workpiece pose from overall workspace images, and a rendering based strategy for synthetic dataset generation was constructed to simulate realistic surface textures of steel workpieces for comprehensive training purposes.

Although the proposed framework realized automatic seam

detection for normal-size steel workpieces in unstructured environments, its performance on large and long steel workpieces may require further improvement. Such workpieces often exceed the field of view of the common RGB cameras and lead to incomplete image information. This can result in erroneous or partial feature extraction, making complicates feature matching and fusion processes. To address these limitations and expand the application of the proposed framework, future research would focus on leveraging more comprehensive datasets and developing feature extraction models capable of capturing localized information.

It is also noted that the proposed framework could currently be less effective for curved or irregular weld seams. The weld seam extraction method proposed in this paper utilizes the underlying assumption that the weld seam forms at the junction of two idealized planes. In real-world steel production, however, the intended weld seams may exhibit distortions or irregularities. Conventional rule-based algorithms may struggle to detect such complex seam geometries. On the other hand, point cloud deep learning models are capable of extracting more sophisticated and nuanced features and offer greater adaptability in a wide range of real-world scenarios. Consideration of specialized deep learning networks could further enhance the ability of the proposed seam detection framework to handle complex weld seams.

Supplementary video of experiments conducted in real production environments to this article can be found online at <https://doi.org/10.1016/j.autcon.2025.106221>.

CRedit authorship contribution statement

Yuzhen He: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Zhaoqi Huang:** Writing – review & editing, Visualization, Conceptualization. **Haotian Liu:** Visualization, Validation & editing, Resources, Funding acquisition. **Xianzhong Zhao:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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